Two Incomparable Denominator Exclusions for $\sum_{n\geq 2}(n!-1)^{-1}$

Finite certificates and exact rationality criteria for Erdős Problem #68

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31 August 2026

Problem 0.1 (Erdős #68)**.** Is

$$S = \sum_{n\geq 2} \frac{1}{n!-1}$$

irrational?

Abstract

Every rational representation $S = a/q$, $q > 0$, of $S = \sum_{n\geq 2}(n!-1)^{-1}$ satisfies

$$q \nmid 299999!, \qquad q \geq 2^{39990} > 10^{12038}.$$

The first exclusion follows from an exact strict-successor carry test; the second from an exact continued-fraction computation. The central difficulty is to retain denominator information after cancellation and then compare it with the positive tail. We exhibit actual prime-power cancellation and show that, at fixed factorial moment, integral channel corrections change the residual only by an integer. The companion constant $S-e+2$ gives an exact factorial-orbit criterion; the remaining assertion is a cofinal miss of the strict factorial successor.

1 The denominator exclusions

Let $S = \sum_{n\geq 2}(n!-1)^{-1}$ and

$$H_m = \sum_{n=2}^{m} \frac{1}{n!-1}, \qquad Z_m = \lfloor m!H_m \rfloor + 1, \qquad b_m = mZ_{m-1} + 1 - Z_m.$$

The divisibility exclusion in the abstract rests on the implication

$$S = a/q, \quad q \mid (m-1)!, \quad m \geq 3 \quad \Longrightarrow \quad b_m = 1. \tag{1}$$

Indeed, $0 < m!(S - H_m) < 1$, by comparison with the telescoping series $\sum_{n>m}(n-1)/n! = 1/m!$. Thus $m!S$ is the unique integer strictly above $m!H_m$. The same argument

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

at $m-1$ gives $Z_m = m!S = mZ_{m-1}$, proving (1). A non-unit carry at index m therefore forces every rational denominator to be at least m, since any smaller positive q would divide $(m-1)!$. The exact value $b_{300000} \neq 1$ therefore excludes every divisor of 299999! as a denominator of S.

The independent continued-fraction exclusion bounds the denominator's size. Neither restriction implies the other: a prime between 299999 and 599998 satisfies the divisibility restriction and fails the size restriction, whereas 299999! does the reverse. Both conclusions are finite; they do not decide the irrationality question posed by Erdős [1, 2]. Their computational certificates are specified in §8.

Adjacent factorial differences affect exactly their divisor channels. This makes the channel equations solvable by integral triangular elimination. Solving them, however, does not make the residual small: at a fixed factorial moment every correction changes it by an integer. Sections 2–4 identify the exact carry test and this limitation of channel cancellation. Section 5 then shows, at the prime 139, why a large common denominator need not survive reduction; Section 6 states the additional real comparison needed.

2 The fixed companion orbit

The termwise identity

$$\frac{1}{n!-1} = \frac{1}{n!} + \frac{1}{n!(n!-1)}$$

replaces the changing prefixes by the factorial orbit of one fixed number,

$$C = \sum_{n\geq 2} \frac{1}{n!(n!-1)}.$$

Absolute convergence gives

$$C + (e-2) = S. \tag{2}$$

The endpoint of the exponential prefix contributes residue 1 modulo m; its positive tail removes another unit on taking the floor. This accounts for the distinguished residue -2 in the following theorem.

Theorem 2.1 (fixed companion-orbit rationality boundary)**.** *The following statements are equivalent:*

1. $S \in \mathbb{Q}$;
2. $(\lfloor m!C \rfloor + 2) \bmod m = 0$ *for every sufficiently large* m.

Consequently,

$$S \notin \mathbb{Q} \quad \Longleftrightarrow \quad (\forall B)(\exists m > B)\ (\lfloor m!C \rfloor + 2) \bmod m \neq 0. \tag{3}$$

The equivalence and its cofinal form are companion orbit boundary.

Proof. First suppose $S = a/q$ with $q > 0$, and take m so large that $q \mid (m-1)!$. Write

$$E_m = m! \sum_{2\leq n\leq m} \frac{1}{n!}, \qquad \varepsilon_m = m! \sum_{n>m} \frac{1}{n!}.$$

Then E_m is an integer, $0 < \varepsilon_m < 1$, and $m!S$ is an integer divisible by m. Multiplying (2) by $m!$ therefore gives

$$\lfloor m!C \rfloor = m!S - E_m - 1.$$

Every summand of E_m except the endpoint $m!/m! = 1$ is divisible by m. Thus $E_m \equiv 1 \pmod m$ and $\lfloor m!C \rfloor \equiv -2 \pmod m$.

Conversely, assume the displayed congruence from some index onward. Let $d_m(C) = \lfloor m!C \rfloor - m\lfloor (m-1)!C \rfloor$ be the canonical factorial digit. Since $0 \le d_m(C) < m$, for $m \ge 3$ the congruence is equivalent to

$$d_m(C) = m - 2.$$

Choose N beyond the exceptional indices. The canonical factorial expansion of C then has the form

$$C = \lfloor C \rfloor + \sum_{m=2}^{N} \frac{d_m(C)}{m!} + \sum_{m>N} \frac{m-2}{m!}.$$

Adding $e - 2 = \sum_{m\ge 2} 1/m!$ and using the telescoping identity

$$\sum_{m>N} \frac{m-1}{m!} = \sum_{m>N} \left(\frac{1}{(m-1)!} - \frac{1}{m!} \right) = \frac{1}{N!}$$

gives

$$S = \lfloor C \rfloor + \sum_{m=2}^{N} \frac{d_m(C)+1}{m!} + \frac{1}{N!},$$

which is rational. Negating the eventual statement yields the cofinal formulation above. □

Remark. The remaining assertion is cofinal escape of this actual factorial orbit from -2. The theorem identifies the required event without supplying it.

2.1 The carry and its wrap correction

The canonical factorial digit is $d_m(C) = \lfloor m!C \rfloor - m\lfloor (m-1)!C \rfloor$. For $C_m = \sum_{n=2}^{m} 1/(n!(n!-1))$, put

$$\delta_m = m!(C - C_m), \qquad \sigma_m = \mathbf{1}_{\{\{m!C\}<\delta_m\}}.$$

The positive tail satisfies $0 < \delta_m < 1/((m+1)! - 1)$. Subtracting it crosses an integer exactly when $\sigma_m = 1$, so for $m \ge 3$

$$b_m = m - 1 - d_m(C) + \sigma_m - m\sigma_{m-1}. \tag{4}$$

This identity is unconditional. A floor-stability hypothesis is required only to discard the two wrap terms. Equality in the defining comparison is the no-wrap case. Canonical factorial expansions and their rationality criterion are classical [3, 4]. For an integer-coefficient factorial series, the standard tail-integrality argument appears in Hančl and Tijdeman [5, Lemma 2.1 and its remark]: a rational sum has integral normalized tails once its denominator divides the preceding factorial. Their subsequent finite-difference criteria impose regularity on the coefficients. In the present series the denominators are $n! - 1$; the companion expansion and the two wrap terms in (4) account for that change.

Theorem 2.2 (exact carry characterisation)**.** *The following conditions are equivalent:*

$$S \notin \mathbb{Q}, \qquad (\forall B)(\exists m > B)\ b_m \neq 1, \qquad (\forall B)(\exists m > B)\ m \nmid Z_m.$$

In particular, cofinal non-unit carries imply irrationality; equivalently, the original problem is the strict-successor criterion.

The pair of equivalences is carry characterisation.

Proof. Rationality forces eventual unit carries by (1). Conversely, if $b_m = 1$ eventually, then $Z_m/m!$ is eventually constant. Since

$$H_m < Z_m/m! \leq H_m + 1/m!,$$

that rational constant is S. Finally writing $x = (m-1)!H_{m-1}$ gives $b_m = m - 1 - \lfloor m\{x\} + 1/(m! - 1)\rfloor$, hence $-1 \leq b_m \leq m - 1$. For $m \geq 3$, these bounds together with $Z_m = mZ_{m-1} + 1 - b_m$, imply $m \mid Z_m$ exactly when $b_m = 1$. ☐

3 An integral basis for factorial channels

We seek integer combinations that cancel selected denominator contributions without cancelling their factorial moment. To keep track of these two requirements, for a finite integer vector λ define

$$W_{d,n} = \frac{n!}{(d!)^{\lfloor n/d \rfloor}}, \qquad M(\lambda) = \sum_n \lambda_n n!, \qquad V_d(\lambda) = \sum_n \lambda_n W_{d,n}.$$

The weights are integers. We first allow the auxiliary coordinate e_1, then impose the manuscript support $n \geq 2$. This coefficient convention introduces no term $1/(1! - 1)$ into S.

Theorem 3.1 (divisor-channel coordinates)**.** *Set*

$$T_n = ne_{n-1} - e_n, \qquad U_n = T_n - \sum_{\substack{d \mid n \\ 2 \leq d < n}} W_{d,n} U_d \quad (n \geq 2).$$

Then

$$M(U_n) = 0, \qquad V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}.$$

The vectors $e_1, U_2, U_3, \ldots$ form an integral basis. Every finite vector has the unique finite expansion

$$\lambda = M(\lambda)e_1 + \sum_{d \geq 2} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} U_d. \tag{5}$$

The observables, the unique expansion and the coefficient formula are divisor channel coordinates.

Proof. The moment of T_n is zero. The floor in $W_{d,n}$ changes between $n-1$ and n precisely when $d \mid n$, so

$$V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d|n}.$$

The recursion cancels the proper divisor channels, proving the two identities by induction. The vectors $e_1, T_2, \dots, T_N$ form an integral triangular basis with final coefficients $1, -1, \dots, -1$. The change from T_n to U_n is integral triangular with unit diagonal. Hence $e_1, U_2, \dots, U_N$ is also an integral basis. Applying M and each V_d identifies its coefficients as in (5). For d beyond the support, $V_d = M$, so no infinite sum is required. □

The prime translator is the case $U_p = T_p$ for a prime $p \geq 3$. The recursion also isolates composite channels; for example

$$U_9 = 9e_8 - e_9 - 5040e_2 + 1680e_3. \tag{6}$$

It has moment zero and only channel 9 is nonzero. Thus each channel can be adjusted separately by a multiple of its own denominator. The expansion is composite translator nine and the observable profile is composite translator nine profile.

3.1 The support restriction and the residual

Write

$$L_D = \mathrm{lcm}_{2\leq d\leq D}(d! - 1), \qquad K_D = L_D e_1 - \sum_{d=2}^{D} \frac{L_D}{d! - 1} U_d.$$

Equation (5) implies both the channel congruence

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1} \tag{7}$$

in the equivalent integer-divisibility form supported channel congruence, and the unique description of every low-channel kernel:

$$V_2 = \cdots = V_D = 0 \quad \Longleftrightarrow \quad \lambda = tK_D + \sum_{n>D} z_n U_n, \qquad M = tL_D. \tag{8}$$

For the full residual

$$\mathcal{R}(\lambda) = \sum_{d\geq 2} \frac{V_d(\lambda)}{d! - 1},$$

we have $\mathcal{R}(e_1) = S$ and $\mathcal{R}(U_n) = 1$. Therefore

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n. \tag{9}$$

The series converges because $V_d = M$ beyond the support. This equation identifies the limit of channel adjustment: every zero-moment correction changes the residual by an integer. A further fractional cancellation coordinate does not arise from these corrections.

Only finitely many z_n are nonzero. Put $a_D = [e_1]K_D$ and $u_n = [e_1]U_n$. Admissibility on $n \geq 2$ becomes the single integer equation

$$ta_D + \sum_{n>D} z_n u_n = 0. \tag{10}$$

Consequently the attainable moments are

$$L_D \frac{g_D}{\gcd(g_D, a_D)} \mathbb{Z}, \qquad g_D = \gcd\{u_n : n > D\}. \tag{11}$$

Indeed, finite integer combinations of the u_n form $g_D\mathbb{Z}$. The resulting positive generator is attained, and an attaining vector is primitive. Odd u_n vanish by induction. For a prime p, the sole nonzero proper-divisor contribution at $2p$ gives $u_{2p} = -2(2p)!/2^p \neq 0$. Every tail thus contains a nonzero coefficient, so $g_D > 0$.

A finite certificate for the infinite moment ideal. The scalar recurrence is

$$u_2 = 2, \qquad u_n = -\sum_{\substack{d\mid n \\ 2\leq d<n}} W_{d,n} u_d \quad (n > 2).$$

Once a divisor g controls the computed tail, only the terms with $d \leq D$ can supply a nonzero residue to a later coefficient. Those finitely many sources have a factorial cutoff:

$$k! \mid W_{d,dk} = \frac{(dk)!}{(d!)^k}.$$

The quotient by $k!$ counts partitions into k unordered blocks of size d.

Theorem 3.2 (finite determination of the moment ideal)**.** *Choose a prime ℓ with $D/2 < \ell \leq D$ and put $H = D(2\ell - 1)$. Then*

$$g_D = \gcd(u_{D+1}, \ldots, u_H), \qquad H < 2D^2.$$

The statement is finite channel moment certificate.

Proof. The finite gcd g divides the nonzero term $u_{2\ell} = -(2\ell)!/2^{\ell-1}$, so $g \mid (2\ell)!$. For $n > H$ and a divisor $d \leq D$, the integer n/d is at least 2ℓ. Hence $g \mid (n/d)! \mid W_{d,n}$. Every early source is divisible by g. Strong induction handles the remaining divisors $D < d < n$, whose coefficients are already divisible by g. Thus g divides the entire tail. Bertrand's postulate supplies ℓ for $D \geq 3$; take $\ell = 2$ for $D = 2$. Finally $H \leq D(2D-1) < 2D^2$. □

Theorem 3.3 (quotient-band breakpoint)**.** *Let λ be a finitely supported integer vector, let $d \geq 2$ and $k \geq 0$ be integers, and suppose each index n in its support satisfies $kd \leq n < (k+1)d$. Then*

$$M(\lambda) = (d!)^k V_d(\lambda).$$

In particular, support in $[d, 2d)$ and $V_d(\lambda) = 0$ force $M(\lambda) = 0$. If all supported indices are at least d, $V_d(\lambda) = 0$ and $M(\lambda) \neq 0$, some supported index is at least $2d$.

Proof. On the stated band $\lfloor n/d \rfloor = k$, so $n! = (d!)^k W_{d,n}$. Sum against λ_n; the final assertion follows by contraposition from the first-band case. □

The exact factorisation is checked for every quotient band at the band identity, and its zero-channel consequence is band cancellation. The first-band form is explicit at first-band factorisation; the final breakpoint alternative is breakpoint witness.

The earlier envelope $\Lambda_n = \operatorname{lcm}(1, \dots, n) \mid u_n$ remains useful for other certificates. Finite-source checks can end sooner because only the sum of the sources has to vanish modulo g. The even-coefficient sign pattern supplies a nonzero term in every tail. Together with (11) this determines the unrestricted minimum positive moment for every fixed depth. For example, $L_4 = 115$, $a_4 = -55$, $u_6 = -180$, $u_8 = -4200$ and $23u_6 - u_8 = 60 \mid \Lambda_9$ give minimum moment 1380, attained by $12K_4 + 253U_6 - 11U_8$. On support through 6 the minimum is instead 4140. Minimum moment and minimum upper support are different objectives. Neither computation supplies a cofinal real-residual gap, and neither alters the two finite denominator exclusions.

4 The moment cost and the choice of support

On the actual support $n \geq 2$, one has the stronger congruence $12 \mid M - 2V_2$. For $n = 2, 3$, $n! = 2W_{2,n}$. For $n \geq 4$, both $n!$ and $2W_{2,n}$ are divisible by 12: writing $n = 2k$ or $2k + 1$, $W_{2,2k} = k! \prod_{j=1}^{k}(2j-1)$ is divisible by 6 for $k \geq 2$. Since every $d! - 1$ is coprime to 6, low-channel annihilation implies

$$12L_D \mid M. \tag{12}$$

The factor 12 is sharp at $D = 2$ and $D = 3$, witnessed respectively by $-6e_2 + e_4$ and $-6e_2 - 8e_3 + 5e_4$.

4.1 Growth of the compulsory factor

The following elementary estimate quantifies this cost:

$$\liminf_{N \to \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3}. \tag{13}$$

The LCM asymptotic is proved here by an ordinary mathematical argument. The linked asymptotic Lean files are proof candidates and are not being cited as kernel verification. The finite-block inequality has a separate Lean source, finite terminal-block inequality. For a terminal block of k terms, the product–lcm–gcd inequality and $\gcd(i! - 1, j! - 1) \mid j!/i! - 1$ give

$$\log L_N \geq \sum_{n=N-k+1}^{N} \log(n! - 1) - \binom{k+1}{3} \log N.$$

Taking $k = \lfloor \alpha\sqrt{N} \rfloor$ gives leading coefficient $\alpha - \alpha^3/6$, maximised by $\alpha = \sqrt{2}$. For a fixed $P \in \mathbb{Z}[X] \setminus \{0\}$, the same ordinary argument gives this bound for $\operatorname{lcm}_{n_0 \leq n \leq N}(n! + P(n))$, where n_0 is sufficiently large that all factors are positive and the collision differences below are nonzero, as in Lai's Lemma 2.1 [7]. The lower cutoff matters: $P = -2$ makes the factor at $n = 2$ vanish. The argument uses

$$\gcd(i! + P(i), j! + P(j)) \mid P(j) - (j!/i!)P(i);$$

the additional $O_P(k^2 \log N)$ loss is lower order. The subtraction identity is recorded by Lai [7, (2.5)]. For $P = 0$, the lcm is only $N!$. These are common-denominator estimates. Consecutive denominators $(N-1)! - 1$ and $N! - 1$ are already coprime for $N \geq 3$, so the qualitative failure of common-denominator clearing follows from two terms; the estimate above is the quantitative growth statement. In particular, $L_N(S - H_N) \to \infty$, so multiplying the prefix by its full lcm cannot make its positive tail small.

4.2 Remote primitive kernels

Support location imposes a different requirement from low moment. Let $D, r \geq 2$ and let L be a positive multiple of $2, \ldots, D$. Put

$$i_j = r + jL \ (0 \leq j < D), \quad N = r + (D-1)L, \quad \alpha_d = (d!)^{L/d},$$

$$H(X) = \prod_{d=2}^{D} (\alpha_d X - 1) = \sum_{j=0}^{D-1} h_j X^j, \qquad A = \prod_{d=2}^{D} \alpha_d.$$

The primitive kernel on this grid is

$$\lambda_j^* = \frac{N! h_j}{A i_j!}, \qquad M = N! \prod_{d=2}^{D} (1 - 1/\alpha_d) > 0.$$

Its final coefficient is 1. The channel equations are polynomial vanishing at the distinct points α_d^{-1}. Integrality follows by expanding h_j/A into products of reciprocal α_d: each product divides the factorial quotient $N!/i_j!$. The minimum step is $L = \operatorname{lcm}(2, \ldots, D)$.

The stronger divisibility

$$r! \prod_{d=2}^{D} (L/d)! \prod_{d=2}^{D} (\alpha_d - 1) \mid M$$

follows by counting partitions into the specified equal-sized blocks. In particular $\lfloor \sqrt{N}/2 \rfloor! \mid M$: if $v = \max(r, L/2)$, then $N < 4v^2$ and $v! \mid M$. Thus these moments absorb every fixed denominator as $N \to \infty$. On the factorial-grid Cramer vector the residual differs from a nonzero determinant times S by an integer: this is the Cramer residual identity. Cofinal residual nonintegrality on this family is sufficient for irrationality; the construction does not establish that nonintegrality.

5 Cancellation in a reduced prefix

A common-denominator bound does not determine the denominator after addition. Put $L_M = \operatorname{lcm}_{2 \leq n \leq M}(n! - 1)$ and $A_M = \sum_{n=2}^{M} L_M/(n! - 1)$. For a prime q with attained maximum $e = \max_{2 \leq n \leq M} v_q(n! - 1) > 0$, set

$$I_{q,e}(M) = \{n \in [2, M] : v_q(n! - 1) = e\}, \qquad R_{q,e}(M) = \sum_{n \in I_{q,e}(M)} \left(\frac{n! - 1}{q^e} \right)^{-1} \pmod{q}.$$

After multiplication by L_M, all lower-valuation terms vanish modulo q. Hence

$$A_M \equiv (L_M/q^e) R_{q,e}(M) \pmod{q},$$

and L_M/q^e is a unit. Consequently

$$v_q(\operatorname{den} H_M) = e \quad \iff \quad R_{q,e}(M) \not\equiv 0 \pmod q. \tag{14}$$

The equivalence, with the maximum and its attainment taken from the actual prefix lcm, is maximal prime power survival. This first-layer test is a consequence of the general reciprocal-sum valuation formula of Louwsma and Martino [6, Lemma 4.1].

The actual factorial-gap examples show why the weights matter. At $M = 138$, the maximal 139-hits are $69, 122, 137$, with cofactor residues $6, 49, 73$, and

$$6^{-1} + 49^{-1} + 73^{-1} \equiv 0 \pmod{139}.$$

At $M = 2592$, the maximal 2593-hits are $349, 2243, 2591$, with residues $1508, 1566, 1678$ and reciprocal sum zero. All these attained exponents are one, so the respective prime disappears completely from the reduced prefix. Singleton maximal support cannot cancel, whereas a hit-count bound alone cannot determine a weighted sum. Even survival must subsequently be coupled to a real strict-successor inequality.

6 One joint collision and residue estimate

The whole-prefix construction separates the common collision factor from prime-power layers with a unique maximal owner. For a natural parameter $p \geq 3$, write $d_n = n! - 1$ and put

$$F_p = (p-1)!, \quad D_p = \operatorname{lcm}_{2 \leq i < j \leq 2p-1} \gcd(d_i, d_j),$$
$$C_p = \operatorname{lcm}(F_p, D_p), \quad L_p^{\mathrm{blk}} = \operatorname{lcm}(F_p, d_2, \dots, d_{2p-1}),$$
$$R_p = L_p^{\mathrm{blk}}/C_p, \quad T_p = \sum_{n=2}^{2p-1} L_p^{\mathrm{blk}}/d_n, \quad \rho_p = (-T_p) \bmod R_p.$$

Every prime dividing R_p has a unique denominator attaining its maximal valuation. Reducing T_p modulo that prime leaves one unit term, giving $\gcd(T_p, R_p) = 1$. Thus $0 < \rho_p < R_p$ whenever $R_p > 1$.

Proposition 6.1 (global complementary-residue criterion). *If arbitrarily large natural parameters $p \geq 3$ satisfy*

$$R_p > 1, \qquad (2p+1)L_p^{\mathrm{blk}} < 2p^2(2p-1)!\rho_p, \tag{15}$$

then S is irrational.

The criterion at natural parameters is global complementary criterion nat.

Proof. Suppose $S = a/q$. For large p, $q \mid F_p \mid C_p$, so $L_p^{\mathrm{blk}} S$ is an integer multiple of R_p. Therefore

$$A = L_p^{\mathrm{blk}}(S - H_{2p-1})$$

is a positive integer congruent to $-T_p$ modulo R_p, and $A \geq \rho_p$. The positive tail bound

$$S - H_{2p-1} < \frac{2p+1}{2p^2(2p-1)!}$$

and (15) give $A < \rho_p$, a contradiction. □

Let $\widetilde{C}_p = C_p/F_p$ and $\eta_p = \rho_p/R_p$. The inequality is exactly

$$\log \widetilde{C}_p - \log \eta_p < \log \frac{2p^2(2p-1)!}{(2p+1)(p-1)!}. \tag{16}$$

The right side is $p \log p + (2\log 2 - 1)p + O(\log p)$. The modulus R_p cancels: increasing private support alone does not improve this comparison. Coprimality permits $\rho_p = 1$, so the complementary gap needs its own quantitative control. Both losses must be controlled at the same parameters. The unit-factor pair floor splits that scale test into the complementary-residue inequality and a local collision-core cap. A joint mean bound below $(1-\varepsilon)p \log p$ on the same nonempty sets in $[X, 2X]$ would suffice; two unrelated cofinal sets can be disjoint.

7 The remaining real comparison

The exact target remains the strict-successor criterion:

$$(\forall B)(\exists m > B)\ m \nmid Z_m. \tag{17}$$

Two sufficient approaches now have explicit, different missing estimates. For the compressed primitive grids, the moments already absorb every fixed denominator. With positive moment M and upper support N, write

$$A_N = \sum_{d=D+1}^{N} \frac{V_d(\lambda)}{d!-1}, \qquad \mathcal{R}(\lambda) = A_N + M \sum_{d>N} \frac{1}{d!-1}.$$

The sufficient strict comparison is

$$\frac{2M}{(N+1)!-1} < \lfloor A_N \rfloor + 1 - A_N. \tag{18}$$

It puts the residual below the next integer and above A_N. Proving this on an unbounded grid family would suffice. The finite signed block A_N is essential. The second approach asks for the simultaneous collision and complementary-gap estimate (16). Neither estimate is established cofinally here.

The limit of short supports. The shortest-support kernels cannot meet (18). For $N = D + O(1)$, the compulsory divisibility $L_D \mid M$ and (13) give

$$\frac{2|M|}{(N+1)!-1} \longrightarrow \infty,$$

whereas the right side of (18) is at most one. A useful family must therefore allow a larger upper support and still control its finite signed block.

Large denominator valuations alone also leave the real gap uncontrolled: the model fractions $1/q^e$ tend to zero. The exact zero-branch / lower-endpoint cylinder and doubled-prime tests require joint numerator or predecessor information. Their thresholds, the adjacent-window collapse, fixed-owner absorption, and the strongest countermodels are retained in the long reasoning record.

8 The finite evidence

An exact non-unit carry at the prime index 67 already forces every rational denominator to be at least 67. The exact GMP carry certificate covers $3 \le m \le 300000$. Its unit carries occur at

$$52,\ 591,\ 1030,\ 1407,\ 1438,\ 2164,\ 4258,\ 10991,\ 21236.$$

In particular $b_{300000} \neq 1$, yielding $q \nmid 299999!$ by (1). This excludes every divisor of that factorial, including large divisors. It does not exclude all integers with small prime factors, since their multiplicities can be too large.

The separate 80000-bit exact rational enclosure forces 23449 common continued-fraction partial quotients. Its best-approximation certificate gives $q \ge 2^{39990} > 10^{12038}$. The open-versus-closed complete-quotient endpoint convention matters for a sharper extraction; no stronger numerical floor is asserted here. The exact sources, enclosure and integer-computation receipts are retained with the long record. Neither finite calculation supplies (17).

Statements and declarations

The finite divisor-event identities, isolated-channel recursion, vanishing moment, individual channel values, and the factor 12 are checked in divisor channel basis. The unique integral basis expansion, attainable-moment formula and compressed primitive-grid construction are ordinary proofs, described in divisor channel basis and, for the grid construction, in compressed primitive channel kernel. The quadratic stopping theorem is checked in complete moment horizon, together with the scalar recurrence, the equal-block divisibility and the prime anchor it uses. The earlier lcm-envelope certificate and depth-4 moment 1380 are ordinary proofs with exact checks in check moment saturation. The finite-source checks are in check finite source. The inherited generic lemmas in tail ideal certificate assume the lcm envelope; they do not formalise the factorial arithmetic. The growth liminf is the ordinary asymptotic consequence of the finite terminal-block inequality cited above. The associated asymptotic and liminf Lean files are proof candidates and are not being cited as kernel verification.
The Comparator isolation pin for the carry equivalences is factorial gap plateau core. factorial zero plateau is the default-root duplicate of that family and is not the isolation source.
Companion-orbit rationality is in companion orbit rationality; the unconditional correction (4) is in companion constant carry law. The pole-residue identity and the two displayed reciprocal equalities are in prime pole criterion. The complete maximal-hit data come from the separate exact modular scan. The carry census and continued-fraction certificate use exact integer computation outside Lean.
The inherited public source pin is `99f4bf47422a`. Source-current additions and that public snapshot must be reconciled at one immutable release before its links are represented as a replay of the whole note. Formal proof checking does not assess external novelty or the significance of the remaining hypothesis. The complete source register, receipts and failed-route calculations belong to the long record.

A Guide to the formal sources

The public `ErdosProblems.Erdos68` package contains the checked source for this note. The pinned snapshot contains fifteen cited modules: `AdjacentUnitCarryWindow`, `CanonicalFactorialDigits`, `ChannelBreakpointRigidity`, `ChannelIntegralCongruence`, `Di-`

`visorFactorialCentre`, `EndpointWeightedPrivateSupport`, `FactorialCarry`, `FactorialChannelCertificate`, `FactorialGapPlateauCore` (isolation pin; `FactorialZeroPlateau` is the default-root duplicate), `FiniteDefectAutomaton`, `GapScalarNormalForm`, `PrimeThresholdParity`, `PrimeUnitTranslator`, `PrimeZeroBranch`, and `StrictSuccessorArithmetic`. Only these public modules belong to the manuscript source surface; no private auxiliary digit-rigidity file is cited or projected. The public root imports every cited module. The declaration table below is pinned to the shared formal-source commit used throughout this problem-note series.

- adjacent-window collapse
- two step den mul transition normalizers
- adjacent unit carry window den eq two step den
- adjacent unit carry window offset eq two step reduced numerator
- denominator at least the carry index
- denominator at least 67
- strict-successor criterion
- cofinal non-unit carries imply irrationality
- zero-branch / lower-endpoint cylinder
- unit-factor pair floor
- Cramer residual identity
- residual centre term
- factorial coeff term
- residual centre
- factorial coeff rat
- factorial eq mul pred factorial
- pred div eq of not dvd
- pred div add one eq of dvd
- residual centre term of not dvd
- residual centre term of dvd
- residual centre term self
- factorial coeff term self
- residual centre recurrence
- weights are integers
- channel weight

- channel weight mul denominator
- channel numerator
- factorial moment
- channel event
- adjacent differences affect only divisor channels
- lambda34
- lambda34 apply three
- lambda34 apply four
- lambda34 channel two
- lambda34 factorial moment
- lambda34 channel three
- lambda34 channel four
- lambda34 channel ge five
- lambda34 residual of tail
- lambda34 residual bounds
- square subsequence radius cubic lower
- no eventual square subsequence cubic upper
- not is little o square subsequence radius
- square subsequence radius three halves lower
- no eventual square subsequence three halves upper

Source-current companion orbit and Comparator routes. The complete infinite rationality boundary is checked in companion orbit rationality. Its paper-facing endpoints are not irrational factorial gap series iff eventually companion floor -2 and irrational factorial gap series iff cofinal companion floor misses. The coherent Comparator composite companion-orbit complete characterisation routes to Theorem 2.1 and Remark 2. The moving-factor Comparator endpoints are recorded in the long reasoning record as the moving-factor scale split, the split-factor normalised collision, and the fixed-owner absorption boundary. These source-current modules, Comparator packages, and this manuscript stage require one common immutable public checkpoint before terminal external replay or Palomar readiness is claimed.

References

[1] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[2] T. F. Bloom, *Erdős Problem #68*. `https://www.erdosproblems.com/68`, accessed 28 July 2026.

[3] G. Cantor, *Über die einfachen Zahlensysteme*, Z. Math. Phys. **14** (1869), 121–128.

[4] J. Galambos, *Representations of Real Numbers by Infinite Series*, Lecture Notes in Math. **502**, Springer, 1976, Chapter 1.

[5] J. Hančl and R. Tijdeman, *On the irrationality of factorial series*, Acta Arith. **118** (2005), 383–401, doi:10.4064/aa118-4-5.

[6] J. Louwsma and J. Martino, *Rational numbers with odd greedy expansion of fixed length*, arXiv:2309.07280v1, 2023, Lemma 4.1, p. 10.

[7] L. Lai, *On the largest prime divisor of* $n! + 1$, arXiv:2103.14894v1, 2021, proof of Lemma 2.4, (2.5).

Companion system context. The claim and trust boundary, cold-clone route to proof authority, and public contribution protocol are described in sibling papers. Those descriptions do not change the mathematical status of this note.